Reproducibility in asset pricing

Advanced Empirical Finance: Topics and Data Science

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What can go wrong doing empirical research?



The Economist, October 2013

Reproducible means: someone else can regenerate every table and figure from the raw data and the code, without asking the authors

Before we ask whether a result is *right*, we ask whether someone else can *recompute* it

This lecture. We look at what breaks reproducibility in practice, run a small research experiment, and build a factor from a portfolio sort step by step

Afterwards you can check a report for reproducibility, explain why a reproducible result still carries researchers' degrees of freedom, and name the choices inside a portfolio sort

Reproducibility in top finance journals

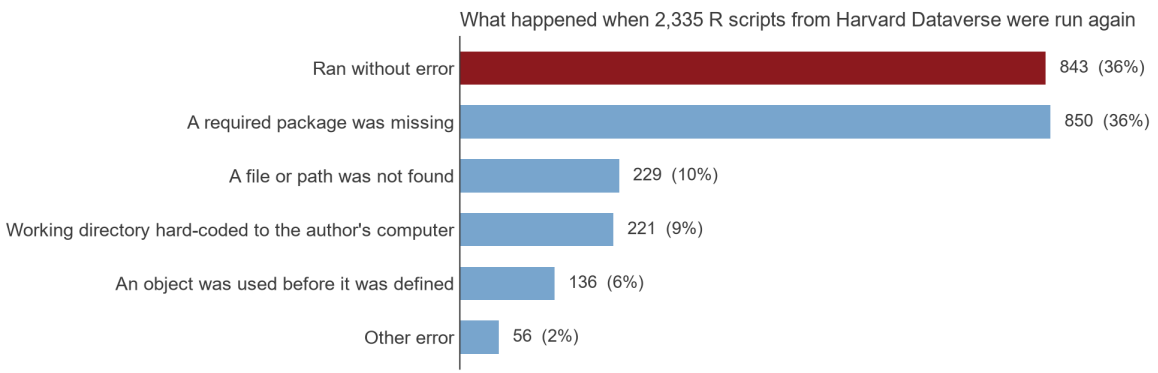


- **Code**: journals require it for empirical work, simulations, numerical computations and experimental studies
- **Data**: if authors cannot share the actual data, they must provide pseudo-data

What makes reproducibility hard?

YOUR TURN **What can go wrong?**

- What problems can occur when running your own project?
- Do you believe your own reports are reproducible?

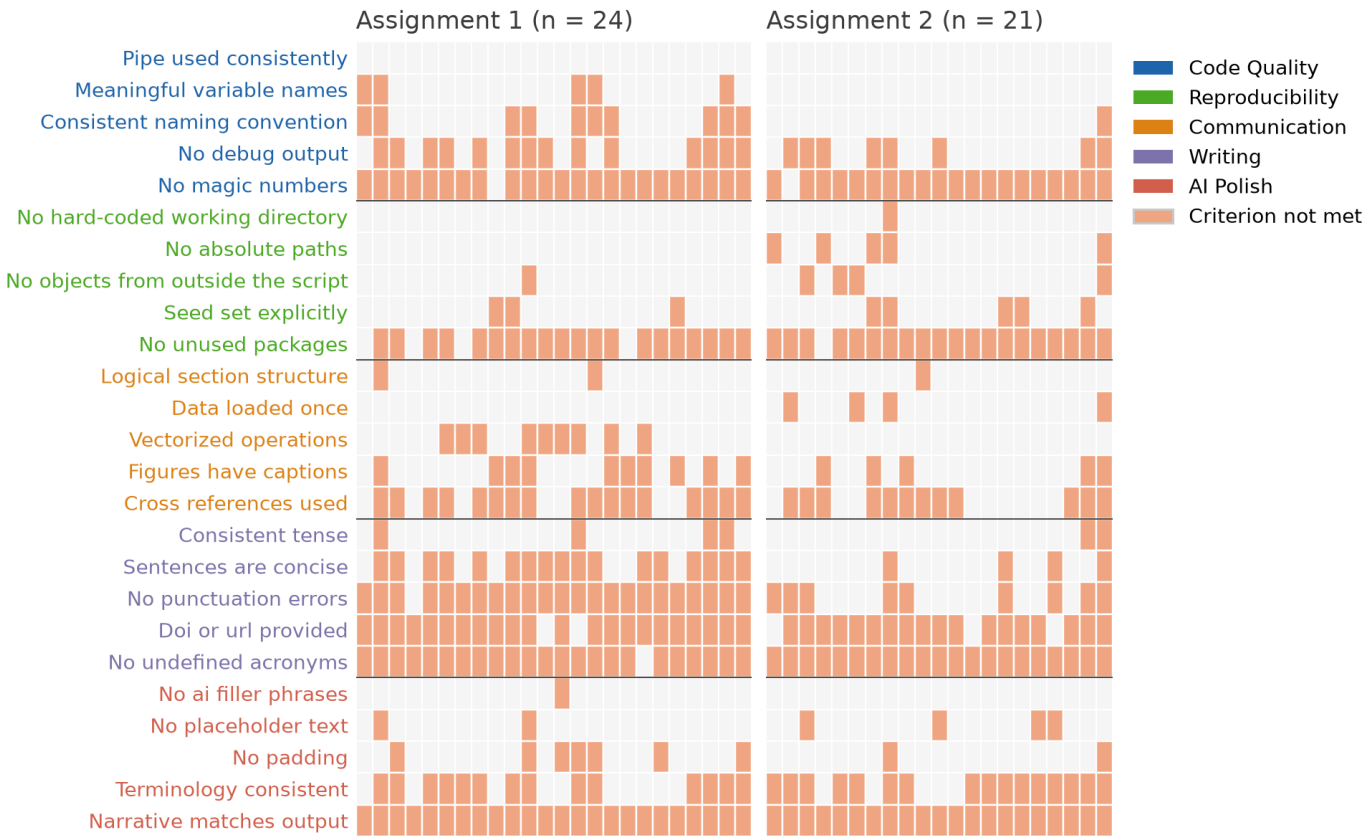


Trisovic et al. (2022), Scientific Data

Minimum requirements

1. Raw data and code that regenerate every table and figure
2. Software and package versions, random seeds
3. Materials openly available, not “upon request”
4. Code that a stranger (= future you) can read

Evidence from this course



- Same checks on this course's two assignments, five representative criteria per group. Unused packages and absolute paths remain the weak spots

Non-standard errors

Suppose we have solved reproducibility. Are we done?

Hello everybody We are starting a large project on market efficiency. Would you join as one of the research teams?

Sure, sounds fun. What do you need from us?

 hf-trading-data.parquet

Hypothesis 1: market efficiency improves over time. Please send me your t-statistic.

 YOUR TURN

1. Data. Daily trading volume of the Dow Jones 30 constituents, 1995 to 2025

R | Python

```r
library(tidyfinance)

dj30 <- download_data(
  domain = "Index Constituents",
  index = "Dow Jones Industrial Average"
)

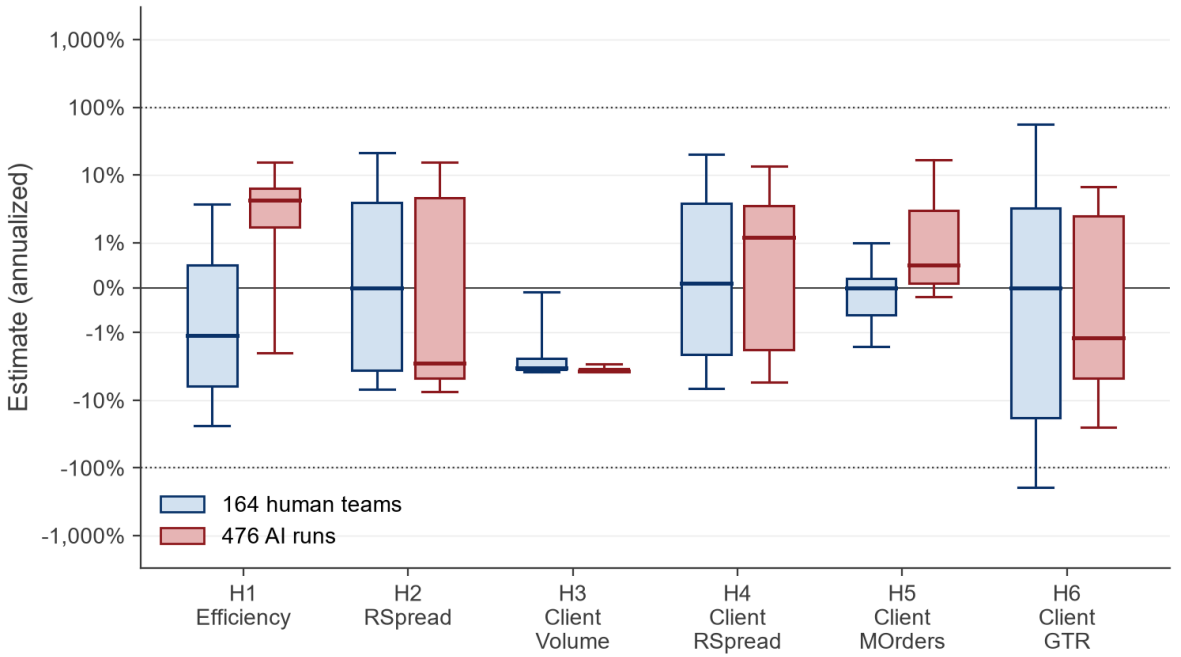
volume <- download_data(
  domain = "Stock Prices",
  symbols = dj30$symbol,
  start_date = "1995-01-01",
  end_date = "2025-12-31"
)
```

2. Hypothesis. Trading volume increased over the last 30 years

3. Send me your t-statistic. One number per person



Non-standard errors



- **164 human teams, same data, same six hypotheses.** Annualized estimates: box is the interquartile range and median, whiskers the 10th to 90th percentile. The dispersion co-varies only weakly with team merits, reproducibility, or peer rating, and participants heavily underestimate it (Menkveld et al., 2024)
- **476 AI runs on the same task** concentrate on a narrow set of analysis paths: less dispersion, but systematically shifted estimates for the complex hypotheses. The choices do not disappear, they move (Huang, Menkveld, and Yu, 2026)

Construction choices in asset pricing

The replication crisis of finance



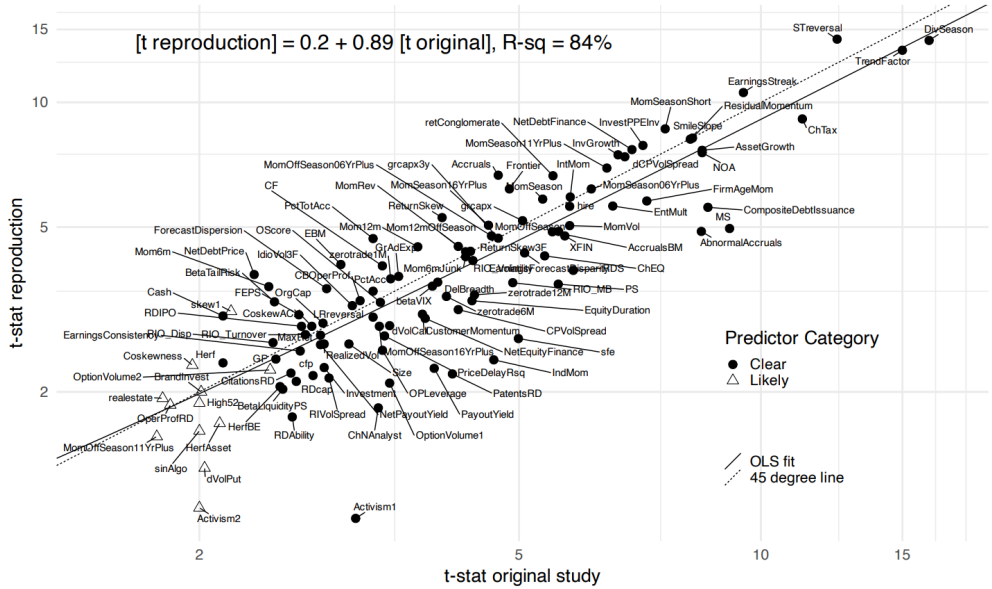
Campbell Harvey
in Harvey, Liu, and Zhu (2016, RFS)

"... most claimed research findings in financial economics are likely false."

"We develop and estimate a Bayesian model of factor replication that leads to different conclusions."



Bryan Kelly
in Jensen, Kelly, Pedersen (2023, JF)



t-statistics of the original studies against the Open Source Asset Pricing reproduction. Source: Chen and Zimmermann (2022), Critical Finance Review; openassetpricing.com

Where do factors come from?

- A stochastic discount factor (SDF) m prices every excess return r_i: $E(m\,r_i) = 0$. Since $E(m\,r_i) = E(m)E(r_i) + \mathrm{Cov}(m, r_i)$,

$$E(r_i) = -\frac{\mathrm{Cov}(m, r_i)}{E(m)} = \underbrace{\frac{\mathrm{Cov}(r_i, -m)}{\mathrm{Var}(m)}}_{\beta_i}\ \underbrace{\frac{\mathrm{Var}(m)}{E(m)}}_{\lambda}$$

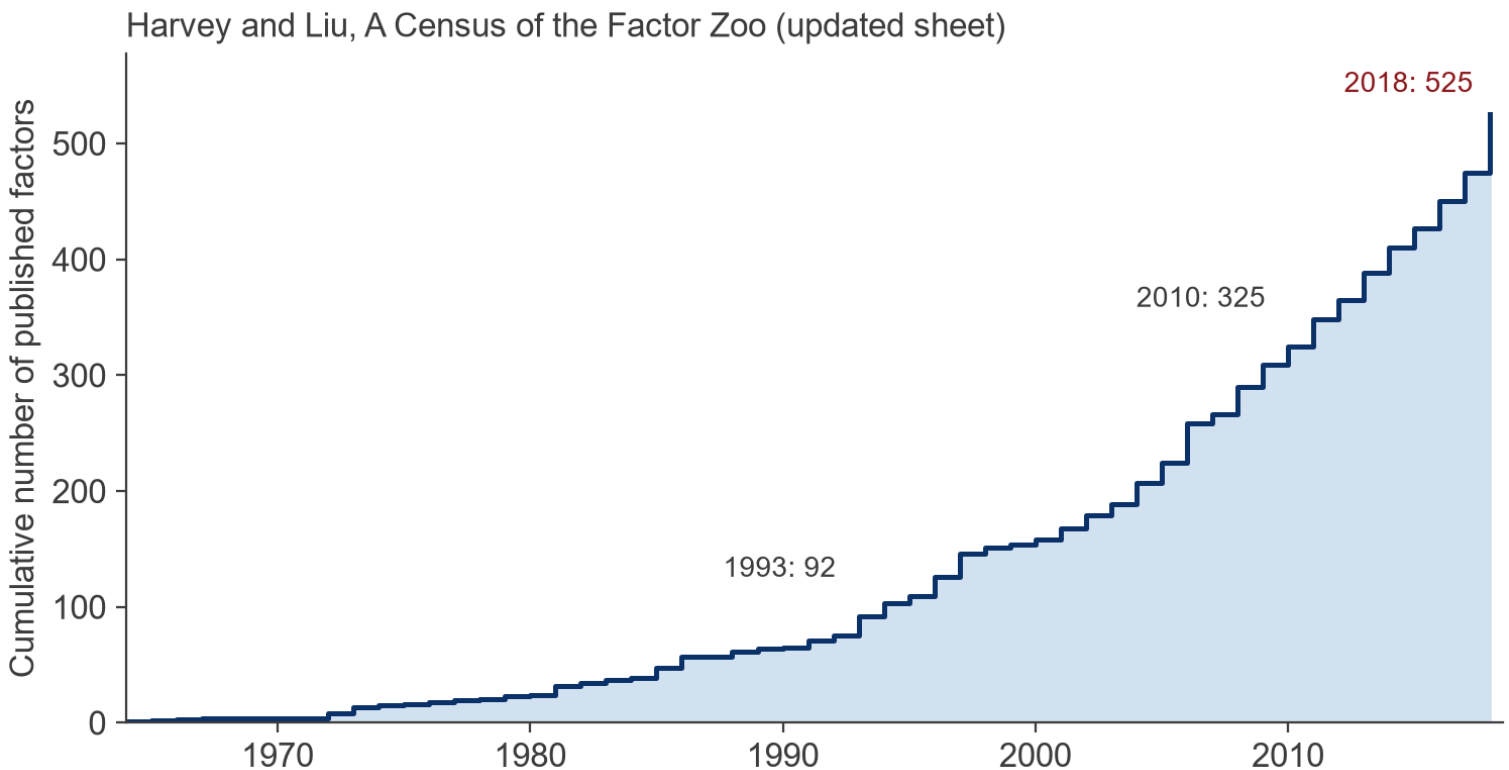
- β_i: the exposure of asset i to bad times, the states in which m is high. $\lambda > 0$: the price of risk, the same for every asset

- We do not observe betas. A characteristic z_i (book-to-market, past returns, size) is useful when it tells us something about the beta

$$E(r_i) = \beta(z_i)\,\lambda$$

- Value firms are often close to distress and lose most in bad states: high beta, high premium

The factor zoo

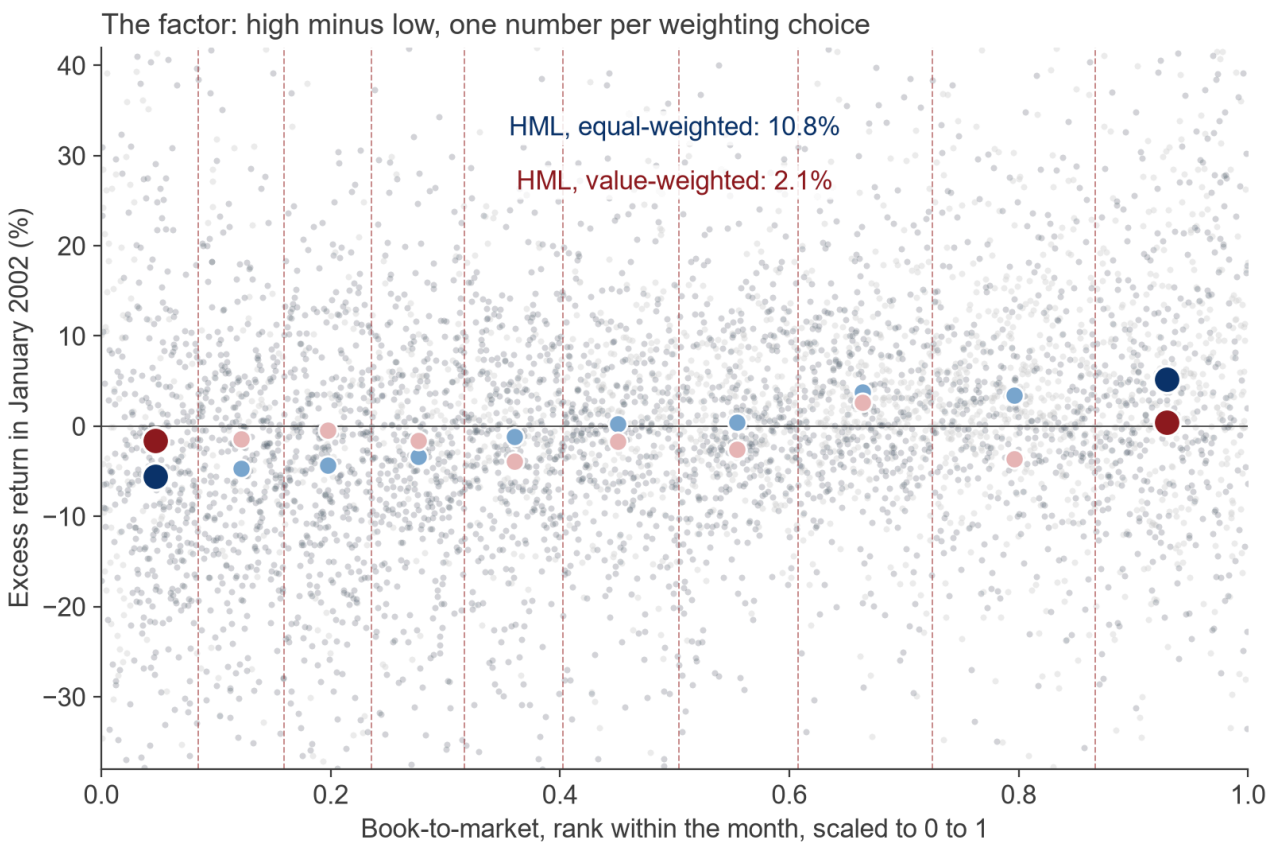


- We can turn any characteristic that predicts returns into a factor. Hence the zoo
- Cochrane (2011): “Now we have a zoo of new factors”
- Do we really believe in 500+ risk factors? See **this sheet** for the census

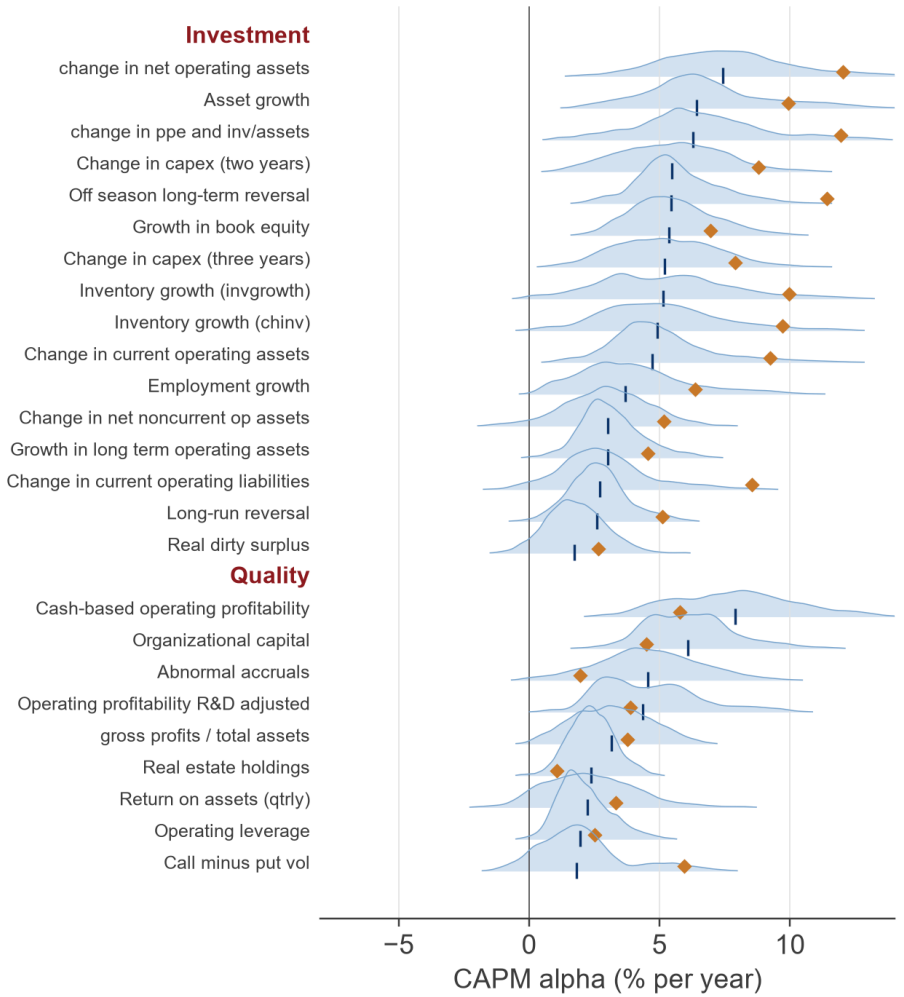
What a portfolio sort does

- Write $E(r_i) = g(z_i) + u_i$, with g the average expected return of stocks with characteristic z
- A **portfolio sort** estimates g without assuming its shape: group stocks with similar z and average their returns
- A **factor** is the top portfolio minus the bottom portfolio

Every step is a choice: which stocks enter, how many portfolios, how we weight stocks, how old the characteristic may be



Construction choices move alphas

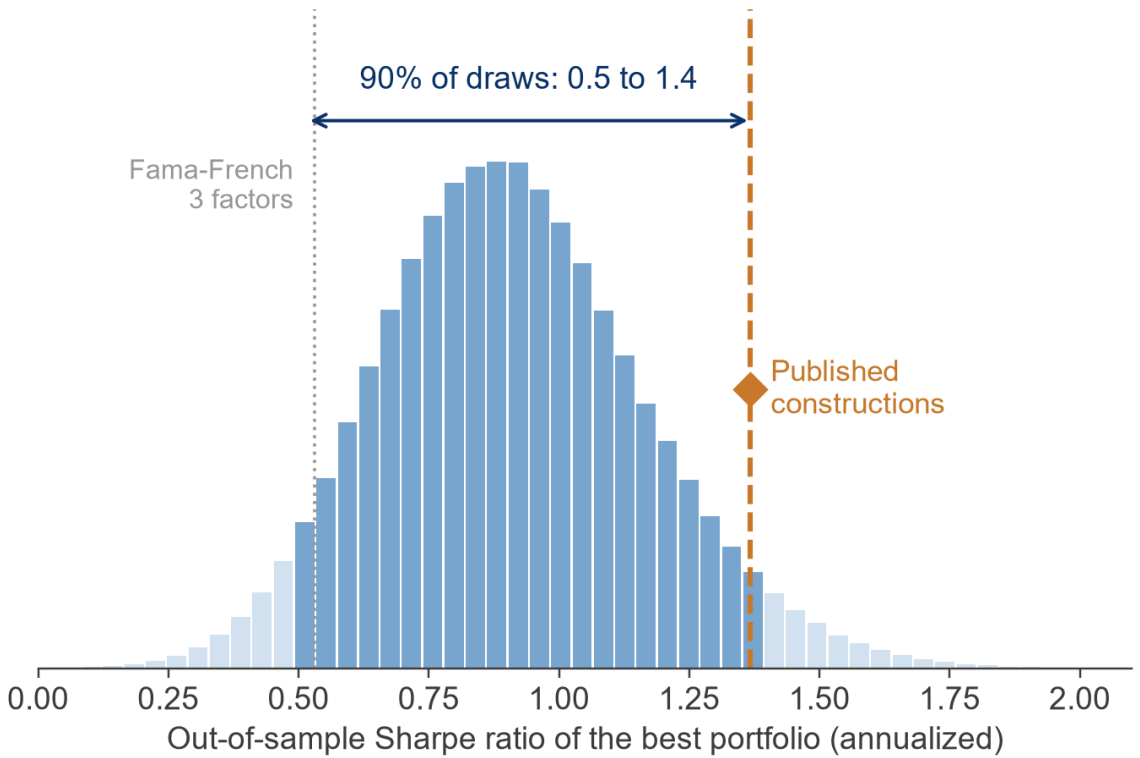


- Voigt and Scheuch (2026) construct 176 concepts from the Open Source Asset Pricing library, each along 23,040 paths. The figure shows two of 13 themes
- Within one concept, the alphas of different paths spread over several percent per year, about as wide as the typical alpha itself. The published path is often far from the median

Distribution of estimated CAPM alphas across all paths of one concept. The tick marks the median path, the diamond the path of the original study.

Data preprocessing choices matter

- Take the market and 50 published factors. Keep the idea of each factor fixed, but build it with one of 3,840 defensible sorting choices
- Draw one choice per factor, compute the out-of-sample Sharpe ratio of the best portfolio, repeat a million times
- Same ideas, Sharpe ratios between 0.5 and 1.4. The published constructions sit at the top: only about 5% of draws do better



Source: Gruber, Kastner, Voigt, and Weiss (2026), *Uncertainty everywhere*

Teaching and research

In this lecture	In the course	In my research
Reproducibility and researchers' degrees of freedom	Every assignment is a Quarto report that regenerates from raw data and code; students make and defend their own modelling decisions	Non-standard errors (Menkveld et al., 2024)
Construction choices in a portfolio sort	Portfolio sorts and backtests on the same CRSP and Compustat data	Which construction choices move factor premia, and by how much (Voigt and Scheuch, 2026)
From one factor to the whole model	Factor models and portfolio choice under estimation error	Pricing the cross-section when the construction of every factor is uncertain (Gruber, Kastner, Voigt, and Weiss, 2026)